Free pythagorean theorem worksheet worksheet 24

Maths | 8th grade | Mixed Operations

1. If 3x - 2y = 4 and x + 4y = 10, what is the value of x + y?

A) 2
B) 3
C) 5
D) 7

2. Evaluate: 2^3 + 3^2 - 5

A) 6
B) 8
C) 12
D) 10

3. Simplify: 3(2a - 4) + 4(a + 1)

A) 10a + 1
B) 10a - 8
C) 6a - 8
D) 10a - 10

4. Find the perimeter of a rectangle with length (2x + 3) and width (x + 5).

A) 6x + 10
B) 6x + 16
C) 8x + 16
D) 8x + 10

5. Factor completely: x^2 - 5x + 6

A) (x - 2)(x - 3)
B) (x - 1)(x - 6)
C) (x - 3)(x + 2)
D) (x + 2)(x - 3)

6. Solve for y: 4y/5 = 8

A) 10
B) 12
C) 15
D) 20

7. Solve for x: 7x - 14 = 21

A) 5
B) 6
C) 7
D) 8

8. What is the value of (3x + 4)^2 when x = 1?

A) 25
B) 49

C) 64
D) 16

9. Which of the following expressions is equivalent to 3(x + 2) - 2(x - 1)?

A) x + 7
B) 5x + 4
C) x + 8
D) x + 4

10. If f(x) = 2x - 5, find f(3) + f(-2).

A) 1
B) -2
C) 3
D) 0

Answer Key

1. C
Solve the system of equations to find x = 4 and y = 1, so x + y = 5.

2. C
Calculate 2^3 = 8 and 3^2 = 9, then 8 + 9 - 5 = 12.

3. B
Distribute to get 6a - 12 + 4a + 4, simplify to 10a - 8.

4. C
Perimeter = 2(length + width) = 2((2x + 3) + (x + 5)) = 8x + 16.

5. A
Find two numbers that multiply to 6 and add to -5, they are -2 and -3.

6. D
Multiply both sides by 5 to get 4y = 40, then divide by 4 to find y = 10.

7. A
Add 14 to both sides to get 7x = 35, then divide by 7 to find x = 5.

8. A
Substitute x = 1 to get (3(1) + 4)^2 = (3 + 4)^2 = 7^2 = 49.

9. C
Distribute to get 3x + 6 - 2x + 2, simplify to x + 8.

10. D
Calculate f(3) = 2(3) - 5 = 1 and f(-2) = 2(-2) - 5 = -9, so 1 + (-9) = -8.